IoT-based Balance and Tremor Exercise Board for Parkinson's with Blood Pressure Monitoring and Android Application

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***Abstract*— Parkinson's Disease (PD) is a neurodegenerative disorder characterized by motor impairments such as tremors, stiffness, postural instability, and slow movements significantly impact a patient's life quality. Traditional clinic-based physiotherapy programs often lacks real-time personalized feedback and remote monitoring capabilities, hence limiting a more consistent long-term care. The proponents proposes an IoT-based Balance and Tremor Exercise Board for Parkinson's Patients. The system utilizes an Raspberry Pi 4 controller, equipped with MPU 6050 sensors for motion and tremor detection, load cells for weight analysis, and an arduino-based blood pressure monitor for vital sign monitoring. Accompanied by an android application for real-time data reading for both patients and medical practitioners. Results from alpha and beta testing shows high reliability, with blood pressure readings maintaining an average of 3.86 mmHg difference from clinical standards. The tremor and balance sensors accurately distinguishing between stable and involuntary tremors. The system successfully provides an affordable, safe, and effective solution for remote rehabilitation, enhancing physical therapies through continuous health monitoring and remote rehabilitations.**

Keywords*— *Parkinson's Disease , Balancing Board, Tremor , Blood Pressure Monitoring , Internet of Things

I. INTRODUCTION

Parkinson's disease (PD) is an irreversible neurodegenerative disorder, and it is a cause of an ever-increasing number of cases mainly in the aging population and later population growth of people with Parkinson's. It is characterized by motor impairments such as tremors, stiffness, slow movements, and instability in the standing position which are, to a large extent, the main factors causing the disease to affect the lives and the independence of millions of people worldwide. Balance disturbances or postural instability, tremors, stiffness, and slow movements in Parkinson's Disease patients are major factors determining life quality and mortality of patients (Yu, 2023). These said symptoms lead to frequent falls and negatively affect the daily life of an individual with PD. Studies show that the effects of bradykinesia, rigidity, impaired proprioception, and freezing of gait are among the primary contributors to balance disturbances in individuals with PD (Park et al., 2015). Various approaches to balance disturbances and/or postural instability have become more available.

The use of Internet of Things (IoT) technologies enhances the monitoring of balance and support in exercises for Parkinson's patients, as the tracking of their posture, stability, and weight distribution is done in real time continuously. The system through IoT-connected sensors placed on the exercise platform collects and analyzes data regarding the patient's balance performance for every session. Thus, the patient's progress can be monitored always, and simultaneously, he/she is supported in keeping the correct posture and improving stability over time. Besides, the use of IoT technology enables the patients to carry out their exercises safely and independently at home, thus reducing the need for occasional clinic supervision and at the same time, being an integral part of daily balance building activities of the patient (Belyaev et al., n.d.).

The traditional in-person therapy methods are limiting for people with Parkinson's, particularly those having issues with balance, because they do not allow constant, personalized, and effective rehabilitation to happen. The patients' physical and mental restrictions that come with the disease further complicate the situation and make it difficult for them to perform daily exercises and remain committed to the therapy. Rehabilitation programs that are based on conventional methods usually do not include real-time monitoring and personalized feedback, which are the reasons why they are not appropriate for remote balance rehabilitation. Indeed, the Internet of Things technologies present brilliant options to tackle these challenges, however, the establishment of specialized systems designed for the different symptoms and stages of Parkinson's, especially for the case of balance dysfunctions, involves a lot of complexity and is costly in terms of time and resources. Hence, there is an urgent requirement for the development of automated, IoT-enabled exercise assistance and monitoring systems for balance rehabilitation that will be able to provide uninterrupted, easily accessible, and tailored support to enhance the patient's functional outcomes and quality of life.

The research will have the following objectives: first create an exercise assistant together with the balance board that uses progressive rehabilitation therapy, sensing the balance, tremors, and blood pressure of the patients, to enhance the safety and efficacy of home exercise for Parkinson's patients with tremors, and impaired coordination. Second, build an android mobile application and a monitoring system that can display real-time data and readings from the prototype that can accommodate both physical therapist and patient; it also monitors the patient's progress. Lastly, to develop data analytics that enables both patients and practitioners to easily understand the data and create

a plan of action and coordination for the decision-making process for the patient.

II. Related Works

Parkinsons Disease is recognized as a rising neurological concern among the Filipino people; there are still a limited number of accessible diagnostic and rehabilitative technologies in the Philippines. Moreover, it was identified a very limited number of local studies focusing on telemedicine, digital rehabilitations, or technology-assisted therapy, especially in rural settings. The author's study highlights the lack of integration of emerging technologies into physiotherapies and physical rehabilitation within Parkinson's Disease management frameworks. This heavily aligns with the goal of the researchers to introduce a localized, accessible, and data-driven rehabilitation system that addresses these technological deficiencies in the Filipino context (Krening et al., 2025) [1].

(Lai et al., 2021) developed a rehabilitation device that focuses on lower-limb exercise by integrating Internet-of-Things and Machine Learning. The system utilizes wearable sensors to gather real-time data during rehabilitation exercises, which are processed using hybrid algorithms such as support vector machines (SVM) and Adaptive Neuro-Fuzzy Inference System (ANFIS) to classify movements and identify exercise types. The study demonstrated an average classification accuracy of over 95%, confirming the feasibility of automatic recognition and evaluation of physical therapy exercises through IoT-assisted frameworks [2].

(Bastardo et al., 2025) conducted a review on existing smart technologies that help PD patients in their diagnosis, rehabilitation, and long-term care. The authors found out that most smart applications aim to improve self-management, symptom monitoring, and therapy adherence through features such as motion tracking, exercise reminders, and progress visualization [3].

A study conducted by Moon, Jung, and Cho (2020) utilized a Wii Fit Balance Board, focusing on the effects of Balance Training using Wii Fit (BTWF). 15 patients with Parkinson Disease were divided into two group of control and experimental, both subjected to traditional occupational therapy for 30 mins per day, 3 times a week, for 8 weeks, while the experimental group have an additional 30 mins of BTWF per session. Both groups showed improvements after intervention, particularly with the experimental group having better overall improvements [4].

This research study is concerned with the design and development of an existing balance board system that incorporates IoT techniques and technologies to provide guided activity based practice for balance, gait and postural stability limitation in persons with Parkinson's disease in the home environment as part of an intervention therapy rehabilitation. This research study addresses the reported concern of Okada et al., (2021) regarding long-term therapy adherence by adapting to the needs of the person using continuous, sensor based systems to support active activity exercise rehabilitation therapy and provide the user with real-time feedback remote support to their home and cumulative archived performance data for the various planned exercise rehabilitation activities [5].

III. Methodology

A) *Conceptual Framework*

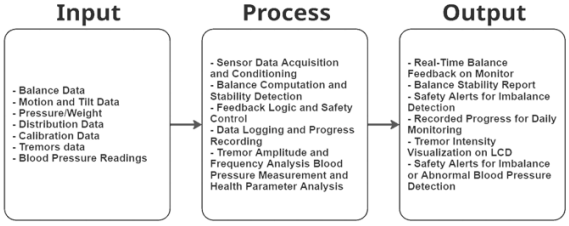


Figure 1: Conceptual Framework

Figure 1. The method of gathering input data includes acquiring data of balance, hand tremor, weight, blood pressure, and manual calibration. The first step of the data collection process is to pass through a series of procedures including acquisition, conditioning, fusion, evaluation, and analysis to extract valuable insights. Among the outputs are the real-time feedback, performance reports, medical professional monitoring, and patient feedback system. The main objectives of the system, which are to support the continuous improvement process, to increase users' involvement considerably and to carry out effective monitoring of both exercises and vital signs, are addressed through these outputs.

B) *Design Methodology*

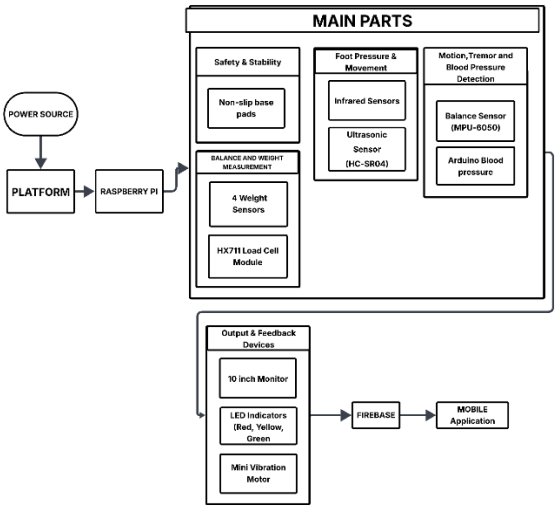


Figure 2: Block Diagram

Figure 2 outlines the flow of the system starting from the power source all the way to the mobile application. The main goal of the system was to ensure and enhance patient balance while, at the same time, monitoring hand tremors and blood pressure continuously. For balance analysis, the platform included load cells, an HX711 module to measure foot pressure and weight distribution. A Raspberry Pi 4 served as the main controller, processing signals from various sensors, including the MPU-6050, which detected motion, tilt, and tremor intensity and frequency. Moreover, the presence of ADXL335, infrared, ultrasonic (HC-SR04), and Arduino-based blood pressure sensor guaranteed that the exercises were stable and safe while taking place and that a continuous check of the patient's vital signs was provided. A 10-inch monitor with LED indicators and vibration feedback displayed the received data, and all the outcomes are transmitted to a mobile app via Firebase for the real-time balance and tremor tracking.

C) System Flowchart

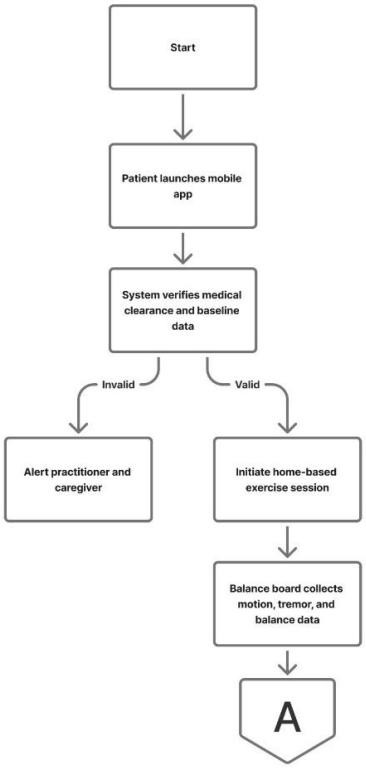


Figure 3: System Flowchart

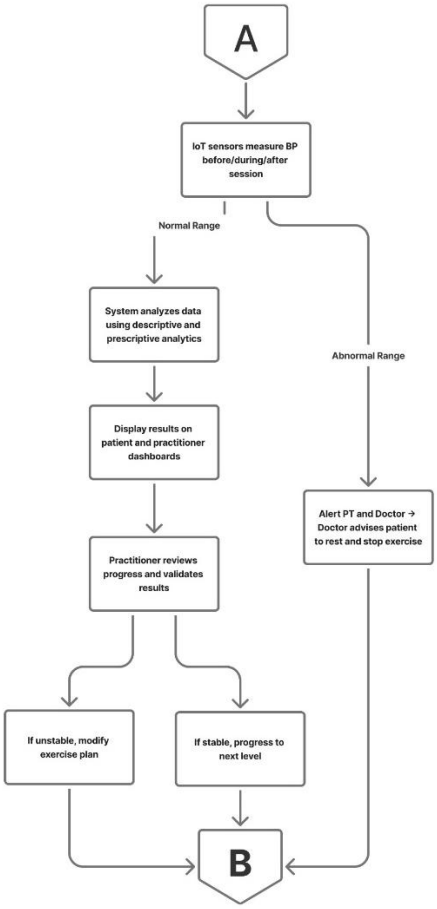


Figure 4: System Flowchart



Figure 5: System Flowchart

The system design and the logical flow of the proposed solution for assisting patients suffering from Parkinson's disease can be described in detail using Figures 3, 4, and 5. Initially, the patient receives permission for a particular workout at home via the mobile application, based on their medical clearance and initial health indicators. Balance board is the key element of the hardware component, providing data on motion, tremors, and balance, which are recorded by the integrated sensors. Together with IoT sensors, it tracks BP parameters. In case any readings are not within the normal range, the system triggers an alert to the PT and asks the patient to stop exercising.

For the cases when the workout was done according to the protocol, descriptive and prescriptive analyses are applied to the collected data, which is later visualized via practitioner dashboards and validated thereupon. In case of unstable results, changes in the plan of exercises should be made or the next level could be set for the patient. Finally, all feedback and new plans for exercise would be delivered to the patient, while the session report would be saved into the database. Notifications would be sent out to both caregiver and practitioner.

IV. Results and Discussions

Table 1: Test 1 Digital vs. Prototype

	Digital BP Monitor	**Prototype BP Monitor**	**Difference**
Before	112/69	115/74	3/5
During	118/70	120/77	2/7
After	119/70	111/73	8/3

Table 1 shows the readings of both the digital blood pressure monitor and the prototype's blood pressure monitor. Table 1 shows that the readings of the prototype BP monitor tracks closely with the digital BP monitor with an average of 4 mmHg difference in the systolic reading and 5 mmHg difference in diastolic reading across the first session. This validates the accuracy of the prototype BP compared to the control digital BP.

Table 2: Test 1 Manual vs. Prototype

	Manual BP (Sphygmomanometer)	Prototype BP Monitor	**Difference**
Before	110/65	115/74	5/9
During	118/70	120/77	2/7
After	119/70	111/73	8/3

Table 2 displays the comparison between the clinical manual blood pressure monitor (sphygmomanometer) and the prototype blood pressure monitor. The data shows that the prototype's reading remains highly consistent with the control manual bp monitor, showing an average difference of 5 mmHg in systolic reading and 6 mmHg in diastolic reading throughout the session. This close alignment further validates the accuracy of the prototype's bp monitor.

Table 3: Test 1 Weight Measurement Comparison

Reading	**Digital weighing scale**	**Hardware scale**	**Difference**
1	52	53	1
2	52	54	2
3	52	55	3
4	52	56	4
5	52	52	0
6	52	54	2
7	52	53	1
8	52	54	2
9	52	55	3
10	52	50	0

Table 3 presents the comparison of readings between a control digital weighing scale and the prototype's hardware scale. The data displays that the reading between the control weighing scale and the prototype's weighing scale are closely aligned with an average difference of 1.8 kg across the first session. This indicates the prototype's capability to provide a consistent and accurate weight measurements relative to a commercial digital weighing scale.

Table 4: Test 1 Average Tremor Sensor Test Result

Test no.	Time Stamp	Left Hand	Right Hand	Remark (Excellent, Good, Fair, Poor)
1	20:51:56 pm	Normal	Normal	Excellent
2	20:51:57 pm	Normal	Normal	Excellent
3	20:51:58 pm	Normal	Normal	Excellent
4	20:51:59 pm	Normal	Normal	Excellent
5	20:52:00 pm	Normal	Normal	Excellent
6	20:52:01 pm	Normal	Normal	Excellent
7	20:52:02 pm	Normal	Normal	Excellent
8	20:52:03 pm	Normal	Normal	Excellent
9	20:52:04 pm	Normal	Normal	Excellent
10	20:52:05 pm	Normal	Normal	Excellent

Table 4 illustrates the results of the average tremor sensor test, conducted over a continuous 10-second interval during the session. The data shows that the prototype successfully identified "Normal" hand stability for both left and right hands throughout the session. Each reading shows an "Excellent" remark, indicating a stable and consistent data without latency or false-positive tremor. This validates the reliability of the MPU-6050 sensors in distinguishing stability from tremors. The classification of the results can be found on table 4.

Table 5: Test 5 Weight Measurement Comparison

Readings	Digital Weighing Scale	Hardware Scale	Difference
1	**67**	**68**	**1**
2	**67**	**68**	**1**
3	**67**	**69**	**1**
4	**67**	**68**	**1**
5	**67**	**68**	**1**
6	**67**	**67**	**0**
7	**67**	**68**	**1**
8	**67**	**68**	**1**
9	**67**	**68**	**1**
10	**67**	**68**	**1**

Table 24 presents the comparison of readings between a control digital weighing scale and the prototype's hardware scale. The data displays that the reading between the control weighing scale and the prototype's weighing scale are closely aligned with an average difference of 1 kg across the first session. This indicates the prototype's capability to provide a consistent and accurate weight measurements relative to a commercial digital weighing scale.

V. Conclusions & Recommendations

A) Conclusions

This study was successfully able to develop and implement a device aimed to provide exercise assistance with the balance board to enhance the safety and efficacy of home exercises for Parkinson's patients with tremors, muscle rigidity, and impaired coordination. The proponents were also successfully able to develop a monitoring system that continuously monitors motion, balance, and tremor over time during the exercise period. Additionally, the proponents were able to implement a monitoring and analytics module that is validated by a physical therapist to provide a clear plan of progression throughout the sessions. The proponents were able to develop a mobile application that provides both patients and physical therapists the data collected from the device. The researchers were able to implement prescriptive and descriptive analytics for data interpretation and data analysis.

The study shows that the device was able to effectively assist in monitoring and improving

balance, detecting tremor activity, reading the weight measurement, and ensuring patient safety through continuous blood pressure tracking, proving its potential as a reliable home-based rehabilitation support system for Parkinson's patients.

B) Recommendation

The developed IoT-based exercise board for Parkinson's patients from the research needs several improvements using considered objectives and data results. For instance, a higher number of patients testing from various stages of Parkinson's disease will yield more accurate data for all stages of Parkinson's. Furthermore, it is recommended to enhance the system's user interface to make it more intuitive and user-friendly, particularly for elderly patients. Improving accessibility features such as clearer instructions and simplified navigation can increase ease of use and patient compliance.

Future improvements may also include integrating data backup and cloud synchronization to ensure secure storage and accessibility of patient records for continuous monitoring.

Lastly, further testing under different conditions and with patients of varying severity levels is encouraged to strengthen the reliability and adaptability of the system in real-world applications.

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